Missionaries and Cannibals

AI-5.py - C:/Users/Vineela/OneDrive/Documents/AI-5.py (3.15.0)

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```
from collections import deque

start = (3, 3, 1)
goal = (0, 0, 0)

moves = [(1, 0), (2, 0), (0, 1), (0, 2), (1, 1)]

queue = deque([(start, [start])])
visited = {start}

while queue:
    (m, c, boat), path = queue.popleft()

    if (m, c, boat) == goal:
        print("Solution:")
        for state in path:
            print(state)
        break

    for dm, dc in moves:
        if boat == 1:
            nm, nc, nb = m - dm, c - dc, 0
        else:
            nm, nc, nb = m + dm, c + dc, 1

        if not (0 <= nm <= 3 and 0 <= nc <= 3):
            continue

        rm, rc = 3 - nm, 3 - nc

        if (nm > 0 and nm < nc) or (rm > 0 and rm < rc):
            continue

        state = (nm, nc, nb)

        if state not in visited:
            visited.add(state)
            queue.append((state, path + [state]))
else:
    print("No solution found.")
```

IDLE Shell 3.15.0

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```
Python 3.15.0 (tags/v3.15.0:d220409, Oct  9 2026, 11:55:56) [MSC v.1951 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/Vineela/OneDrive/Documents/AI-5.py ==========
Solution:
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)
>>>
```